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DETAIL SPECIFICATION

ACTUATOR, ROTARY AND LINEAR, HYDRAULIC

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for material, design, fabrication, and testing of rack-and-pinion hydraulic rotary actuators and hydraulic linear actuators for naval shipboard use. The primary application for the actuators is in remote operation of various ship systems' components.

1.2 Classification. Hydraulic actuators are of the following types (see 6.2).

1.2.1 Rotary. Rotary actuators convert hydraulic power to a rotational output.

1.2.2 Linear. Linear actuators convert hydraulic power to a linear output.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-PRF-6083 - Hydraulic Fluid, Petroleum Base, for Preservation and Operation

MIL-PRF-8625 - Anodic Coatings for Aluminum and Aluminum Alloy

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-PRF-8805 - Switches and Switch Assemblies, Sensitive, Snap Acton (Basic, Limit, Push Button and Toggle Switches), General Specification for

MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service

MIL-PRF-17672 - Hydraulic Fluid, Petroleum, Inhibited

MIL-DTL-24441 - Paint, Epoxy-Polyamide, General Specification for

MIL-DTL-32353 - Hydraulic & Lubricating Oil, Synthetic Hydrocarbon Base

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

MIL-STD-2193 - Ship Hydraulic System Components

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 2001-D04 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth

(Copies of this document are available online at www.agma.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available online at www.asme.org.)

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ASTM INTERNATIONAL

ASTM F718 - Standard Specification for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet

(Copies of this document are available online at www.astm.org. Copies of NAVSEA-reviewed data sheets are available online at www.nstcenter.biz.)

SAE INTERNATIONAL

SAE AS4059 - Aerospace Fluid Power - Contamination Classification for Hydraulic Fluids

SAE AS4716 - Gland Design, O-ring and Other Seals

(Copies of these documents are available online at www.sae.org.)

SSPC: THE SOCIETY FOR PROTECTIVE COATINGS

SSPC-SP1 - Solvent Cleaning

(Copies of this document are available online at http://www.sspc.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), at least one actuator of each specific part number shall be subjected to all first article inspections of table I in accordance with 4.2.

3.2 Materials and fabrication. Materials and fabrication shall be in accordance with MIL-STD-2193. Materials shall be compatible with the fluid, temperature, service, and performance requirements specified herein. Direct contact of electrolytically dissimilar metals shall be avoided to the maximum extent possible, and are not permissible for any actuator designs that will be partially or completely submerged in seawater or exposed to conditions conducive for galvanic corrosion.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Prohibited materials. The actuators shall not contain any materials or chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.3 Design.

3.3.1 Pressure requirements.

3.3.1.1 Rated operating pressure. Unless otherwise specified (see 6.2), actuators shall be designed for a rated operating pressure of 3,000 psig (lb/in^2 gauge) (20.7 megapascal [mPa]).

3.3.1.2 Proof pressure. Actuators shall withstand a proof pressure no less than 1.5 times the rated operating pressure without permanent deformation or external leakage sufficient to form a drop. For unbalanced actuators, the proof pressure for ports subject to pressure multiplication shall be not less than the rated operating pressure times the multiplication factor. Pressure containing parts shall be designed such that they do not experience a stress in excess of 90 percent of the minimum specified yield strength when subjected to proof pressure.

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3.3.1.3 Burst pressure. Actuators shall be designed such that no part of the external pressure boundary ruptures due to excessive material deflection or failure when subjected to a burst pressure of 2.5 times the rated operating pressure or 1.5 times the proof pressure, whichever is greater.

3.3.2 Fluid compatibility. Actuators shall be compatible with the specified end-use hydraulic system fluid (see 6.2). Unless otherwise specified (see 6.2), actuators shall meet the requirements specified herein while operating with any hydraulic fluid in accordance with MIL-DTL-32353, MIL-PRF-17331, or MIL-PRF-17672. Actuators intended for use in systems with any hydrocarbon base fluid shall also be compatible with preservative fluid in accordance with MIL-PRF-6083.

3.3.3 Envelope. Dimensions shall be within the envelope requirements as specified (see 6.2).

3.3.4 Orientation. Actuators shall operate in any orientation or position. Actuator mounting bolt and dowel pin patterns shall meet the requirements specified (see 3.4.12).

3.3.5 Rotation. The shaft rotating angle for a rotary actuator and its tolerance shall be as specified (see 6.2).

3.3.6 Operating cycles. Unless otherwise specified (see 6.2), actuators shall be designed to achieve a minimum operating life of 50,000 cycles. For linear actuators and two-position rotary actuators, a cycle shall be defined as actuating from one extreme position to the other and return. For three-position rotary actuators, a cycle shall be defined as the shaft rotating from the neutral position to one extreme position and back to neutral, followed by rotation to the other extreme position and return to neutral.

3.3.7 Actuator output. Minimum and maximum actuator output at the required pressures shall be as specified (see 6.2). Rotary actuator output shall be as specified in units of torque whereas linear actuator output shall be as specified in units of force. The minimum actuator output required shall be obtained at a differential pressure not more than two-thirds the rated operating pressure. For rotary actuators, the maximum actuator output developed at a differential pressure equal to the rated operating pressure shall not exceed the minimum actuator output specified by more than 200 percent.

3.3.8 Breakaway pressure. Differential pressure required to start motion of the shaft in either direction under no-load conditions shall not exceed 100 psi (0.7 mPa). Actuators shall comply with this requirement after being held in either extreme position for at least 1 minute at the rated operating pressure.

3.3.9 Dashpots. When specified (see 6.2), dashpots shall be designed to decelerate the motion of the piston near the end of the stroke. Dashpot design shall be determined by the manufacturer to satisfy the specified performance requirements (see 6.2).

3.3.10 Leakage. Unless otherwise specified (see 6.2), internal leakage shall not exceed 0.003 ounce (0.1 cubic centimeter) per 5 minutes at differential pressures up to the rated operating pressure. External leakage past static seals shall be insufficient to form a drop under all loading conditions. External leakage past dynamic seals shall not exceed 1 drop per inch of diameter or fraction thereof for every 25 cycles under all loading conditions.

3.3.11 Seals. Unless otherwise specified (see 6.2), seals, packings, seal glands, and back-up rings shall be in accordance with MIL-STD-2193. Non-metallic seals, packings, and back-up rings shall be compatible with the end-use hydraulic system fluid (see 3.3.2).

3.3.12 Noise. When required (see 6.2), actuators shall not exceed the allowable structureborne noise limits specified (see 6.2).

3.4 Construction.

3.4.1 Actuator bodies. Actuator bodies shall comply with the general construction requirements of MIL-STD-2193. External pressure boundary parts shall be designed to a minimum factor of safety of one using one-quarter of the minimum specified tensile strength or two-thirds of the minimum specified yield strength of the material, whichever is lower, when subjected to 110 percent of the rated operating pressure (see 3.3.1.1).

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3.4.2 Aluminum bodies. Rotary actuators with an output greater than 1,200 inch-pounds (in-lb) (136 newton meters [nm]) at rated operating pressure that utilize aluminum as the body material shall be furnished with a corrosion-resistant replaceable sleeve to separate the piston from the aluminum body to prevent galling, corrosion, seizing, or excessive wear of the aluminum body. Exterior surfaces of actuator aluminum bodies shall be anodized in accordance with MIL-PRF-8625 types IC, II, IIB, or III, or painted as specified in 3.6.3.

3.4.3 Welding and brazing. Brazing, when permitted (see 6.2), shall not be used in fabrication of actuator parts subject to hydraulic pressure. Welding, when permitted (see 6.2), shall be in accordance with S9074-AR-GIB-010/278. Weld repair of base materials is not permitted.

3.4.4 Chamfers. Unless otherwise specified (see 6.2), methods of chamfering for seal lead-in shall be in accordance with SAE AS4716.

3.4.5 Actuator ports. Unless otherwise specified (see 6.2), actuator ports shall comply with the fluid connection requirements of MIL-STD-2193. Tapered pipe threads such as NPT are not permitted.

3.4.6 Fluid porting. Unless otherwise specified (see 6.2), transfer of hydraulic fluids across the actuator shall be by internal porting within the actuator and shall be sized for fluid velocities at rated flow no greater than 20 feet per second.

3.4.7 Fasteners. Fasteners shall be in accordance with the requirements of MIL-STD-2193. The minimum acceptable thread engagement for fasteners shall be that which develops the ultimate tensile strength of the fastener prior to stripping either the internal or external threads as calculated by the formulas of FED-STD-H28. Fastener and hole chamfers as well as bolted part tolerances shall be factored into required length of engagement for bolted joints.

3.4.8 Bearings. The maximum design bearing loads shall not exceed the basic dynamic capacity rating (capacity for one million revolutions) of the bearing used.

3.4.9 Gear chamber. Gear chambers for rotary actuators shall be fully enclosed and designed to withstand rated operating pressure and proof pressure. Unless otherwise specified (see 6.2), the gear chamber shall be filled with end-use hydraulic system fluid (see 3.3.2) and be designed to operate when filled ¾ full of fluid to allow for thermal expansion.

3.4.9.1 Gearing. Only spur gearing shall be used with the design in accordance with AGMA 2001-D04. Pinion gear and rack shall be match-marked for proper assembly. The material for the pinion shall be corrosion resistant. The spline fit shall be as specified (see 6.2). The rack-and-pinion gear design shall limit the bending stresses in the teeth to 65 percent of the material yield strength when the output shaft is stalled and the actuator pressurized to a differential of 110 percent of the rated operating pressure across the rack.

3.4.9.2 Relief valves. Unless otherwise specified (see 6.2), gear chambers for rotary actuators shall be equipped with relief valve or other pressure-limiting device to ensure the pressure in the gear chamber does not exceed the specified proof pressure under any mode of operation. The operating requirements of the relief valve shall be as specified (see 6.2).

3.4.10 Position indication. Rotary actuators shall be complete with visual position indication. Visual position indication shall consist of markings stamped or engraved in an area adjacent to the driven component's openings (such as switch covers or actuator bodies) and a pointer stamped or engraved in or attached to the driven component. Position identification markings shall be provided as specified (see 6.2).

3.4.10.1 Electrical position indication. When specified (see 6.2), remote electrical position indication shall be provided as part of the actuator assembly. Remote electrical position indication shall consist of sensitive switches for mounting on or adjacent to the actuator. Moving parts shall be fitted with covers or guards to prevent equipment damage and personnel injury. Unless otherwise specified (see 6.2), sensitive switches shall be in accordance with MIL-PRF-8805.

3.4.11 Shims. Shims may be used as required to achieve position accuracy. Adjustments due to shimming shall not exceed 2 degrees for rotary actuators. If shimming is used, allowances shall be made for minimum thread engagement (see 3.4.7). Non-metallic shims shall be installed in actuators where galvanic action may occur between the shims and surrounding material.

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3.4.12 Mounting. The mounting interface configuration, to include mounting bolt and dowel pin patterns, spline diameter and missing tooth location (rotary actuators only), and piston rod diameter and mating interface (linear actuators only), shall be as specified (see 6.2).

3.4.13 Castings and cast materials. Castings and cast materials shall be inspected to a non-destructive test procedure in accordance with S9074-AR-GIB-010/278. Pressure containing castings shall be inspected in accordance with the table for category 2, pressure containing castings – machinery or pressure vessel castings, and non-pressure containing castings shall be inspected in accordance with the table for category 1, non-pressure containing castings in machinery or pressure vessels. Repair of castings and cast materials is not permitted.

3.5 Environmental conditions. The environmental conditions governing actuator design shall be as follows:

3.5.1 Ambient pressure and temperature. Unless otherwise specified (see 6.2), actuators shall be designed to operate continuously under the ambient pressure and temperature ranges provided in MIL-STD-2193.

3.5.2 Vibration. Actuators shall be designed to withstand external shipboard environmental vibration in accordance with MIL-STD-167-1, type I, without damage to or loosening of parts, or degradation of operational performance, including leakage, below specification requirements.

3.5.3 Shock. Actuators shall be designed such that no part exceeds its material yield strength when subjected to shock and shall be qualified in accordance with MIL-DTL-901 for the specified grade and class (see 6.2).

3.5.4 Salt fog. Actuators shall withstand exposure to a salt-laden environment when tested in accordance with 4.5.10 without incurring corrosive damage or degradation of operational performance below specification requirements.

3.6 Material finishes. Surface roughness shall be determined in accordance with MIL-STD-2193 and ASME B46.1 and shall be as follows.

3.6.1 External finishes. External surface finishes shall not exceed 250 microinches (Ra), except castings, which shall be as cast. External surfaces shall be smooth, free of burrs, sharp edges, and other irregularities.

3.6.2 Internal finishes. Internal surface finishes shall be as follows:

a. Internal cylinder bore or piston rod over which packings or seals must slide shall be not less than 8 Ra nor greater than 16 Ra.

b. Internal moving parts in wearing contact other than as specified in 3.6.2.a shall not exceed 32 Ra.

c. Internal non-moving parts other than parts designed for applications specified in (a) and (b) shall not exceed 125 Ra.

d. Finished surfaces for either dynamic or static packing or seal grooves shall be in accordance with the specifications identified in MIL-STD-2193.

3.6.3 Coatings. External surfaces of non-corrosion resistant metal, with the exception of threaded areas or machined surfaces in rolling or sliding contact, shall be cleaned, prepared, and coated with a two-coat paint system in accordance with MIL-DTL-24441 as follows:

a. Grease, oil, and dirt shall be removed from external surfaces by solvent wiping, vapor decreasing, or caustic washing and rinsing. This does not apply to aluminum parts immediately after anodizing. Solvents and other cleaning products shall first be verified for compatibility with the base material prior to use. Cleanliness of external surfaces shall be verified in accordance with SSPC-SP1.

b. Once verified clean, surfaces shall be prepared using appropriate methods to be determined by the manufacturer to ensure adhesion of the applied paint system.

c. Surfaces shall first be painted with an epoxy polyamide primer coat in accordance with MIL-DTL-24441, type IV (Formula 150) to a minimum dry film thickness of 4 mils. Primed surfaces shall then be painted with a dark gray epoxy polyamide top coat in accordance with MIL-DTL-24441, type IV (Formula 151) to a minimum dry film thickness of 4 mils. The total minimum dry film thickness shall be 8 mils. Both primer and top coats shall be applied in accordance with the applicable NAVSEA-reviewed ASTM F718 data sheet with sufficient cure time allowed between coats.

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3.7 Nameplates. Nameplates shall be provided and mounted on each actuator. Unless otherwise specified (see 6.2), the nameplate shall be in accordance with the identification requirements of MIL-STD-2193.

3.8 Maintainability. Actuators shall be designed for ease of servicing and maintenance. The Mean Time to Repair (MTTR) for normal corrective maintenance shall not exceed two hours. At a minimum, normal corrective maintenance shall consist of actuator disassembly, replacement of seals, backup rings, and any parts or subassemblies intended to be carried as onboard repair parts, and actuator reassembly.

3.8.1 Special tools. No special tools, fixtures, or wrenches shall be required for disassembly, service, or maintenance of the actuators. Special tools are defined as those tools not listed in the GSA Global Supply Catalog.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of the tests specified in table I. The sequence of the tests shall be group A, followed by group B, and then group C (see 6.3).

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TABLE I. First article inspection.

Examination or test	Requirement	Test method
Group A		
Visual examination	3.2, 3.3, 3.4, 3.6, 3.7	4.5.1
Group B		
Run-in	3.3.6, 3.3.10, 3.4.10.1	4.5.2
Proof pressure	3.3.1.2, 3.3.10	4.5.3
Breakaway pressure	3.3.8	4.5.4
Actuator output	3.3.7	4.5.5
Low pressure cycling	3.3.6, 3.3.8	4.5.6
Relief valve operation (as required)	3.4.9.2	4.5.7
Position accuracy (rotary actuators only)	3.3.5	4.5.8
Noise	3.3.12	4.5.9
Dashpot performance (as required)	3.3.6, 3.3.9	4.5.10
Maintainability	3.8	4.5.11
Group C		
Vibration	3.5.2	4.5.12
Shock	3.5.3	4.5.13
Endurance	3.3.6	4.5.14
Salt fog	3.5.4	4.5.15
Burst pressure	3.3.1.3	4.5.16

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests listed in table II. Each actuator shall be subjected to conformance inspection.

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TABLE II. Conformance inspection.

Examination or test	Requirement	Test method
Visual examination	3.2, 3.3, 3.4, 3.6, 3.7	4.5.1
Run-in	3.3.6	4.5.2
Proof pressure	3.3.1.2, 3.3.10	4.5.3
Breakaway pressure	3.3.8	4.5.4
Actuator output	3.3.7	4.5.5
Low pressure cycling	3.3.6, 3.3.8	4.5.6
Relief valve operation (as required)	3.4.9.2	4.5.7
Position accuracy (rotary actuators only)	3.3.5	4.5.8
Noise	3.3.12	4.5.10

4.4 Test conditions. Test conditions shall be as specified herein.

4.4.1 Test fluid. Unless otherwise specified (see 6.2), the test fluid shall be the end-use hydraulic fluid (see 6.2) or one of the fluids specified in 3.3.2.

4.4.2 Ambient test conditions. Unless otherwise specified (see 6.2), actuators shall be tested under ambient temperatures between 59 to 104 degrees Fahrenheit (°F) (15 to 40 degrees Celsius [°C]).

4.5 Test methods.

4.5.1 Visual examination. The actuator shall be visually and dimensionally examined to verify conformance to this specification. The list of materials used shall be inspected to verify that prohibited materials have not been used (see 3.2.2). Deviation from the specified dimensions, materials, finishes, and configuration shall be cause for rejection.

4.5.2 Run-in test. The actuator shall be cycled (see 3.3.6) at least ten times at the rated operating pressure to ensure the chambers are properly filled with hydraulic fluid and that any trapped air is purged from the actuator. During this test, actuators shall not exhibit any binding, instability, unusual noise (e.g., squealing, grinding, or chattering), change in performance, or leakage exceeding the limits specified (see 3.3.10). If the actuator is equipped with electrical position indication switches (see 3.4.10.1), the switches shall be continually tested to ensure that they are functioning in response to changes in actuator position.

4.5.3 Proof pressure test. Each actuator port shall be pressurized to the proof pressure specified in 3.3.1.2 for a minimum of 15 minutes. When pressurizing each port, the actuator shall be permitted to travel to its full stroke. Leakage exceeding the allowable limits specified in 3.3.10 or any signs of loosening, damage, or permanent deformation shall be cause for rejection.

4.5.4 Breakaway pressure test. The breakaway pressure shall be obtained by first holding the actuator in one of its extreme positions under rated operating pressure for a minimum of 24 hours for first article inspection or 15 minutes for conformance inspection. The actuator shall then be shifted to the opposite position while observing the differential pressure required to start movement. The test shall be repeated by shifting the actuator in the opposite direction. Breakaway pressure in excess of the specified value (see 3.3.8) shall constitute a failure of the test.

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4.5.5 Actuator output test. Actuators shall meet the minimum and maximum output limits specified (see 3.3.7) when measured and recorded a minimum two times at each applicable test pressure condition. The actuator must be depressurized and returned to the original position prior to measuring actuator output at each test pressure. There shall be no evidence of loosening, binding, or damage to the actuator shaft or parts. Temperature of the test fluid shall be maintained at between 68 and 140 °F (20 and 60 °C) while performing the actuator output test.

4.5.6 Low pressure cycling test. Actuators shall be continuously cycled (see 3.3.6) at least ten times by slowly applying pressure to the respective actuator ports. For each actuator stroke, the initial breakaway and average running pressures shall be recorded. Evidence of binding, breakaway pressure exceeding the specified value (see 3.3.8), or pressure spikes during each actuator stroke that exceed the breakaway pressure shall be cause for rejection.

4.5.7 Relief valve operation test. Rotary actuators equipped relief valves or other pressure-limiting devices shall be subject to a relief valve operational test. Pressure shall be slowly applied to the relief valve until the relief valve unseats. Relief valve cracking pressure shall be recorded. A cracking pressure greater than the maximum allowable relief valve cracking pressure (see 3.4.9.2) is cause for rejection.

4.5.8 Position accuracy test. Rotary actuators shall be cycled for a minimum of three complete cycles from one extreme position to the other and back to the original position. The maximum rotation shall be measured and verified to fall within the shaft rotating angle and tolerance specified (see 3.3.5).

4.5.9 Noise test. Actuators shall meet the structureborne noise as specified in 3.3.12. Unless otherwise specified (see 6.2), actuators shall be noise tested with a nominal supply pressure of 2,800 psig, a nominal return pressure of 200 psig, and a test fluid viscosity between 180 and 50 centistokes. Actuators shall meet the specified noise requirements (see 3.3.12) for the full range of motion at rated load and speed. Rotary actuators shall be tested during rotation to each position separately. Linear actuators shall be tested from the fully retracted position to the fully extended position and separately from the fully extended to the fully retracted position.

4.5.10 Dashpot performance test. Actuators equipped with dashpots shall be cycled for a minimum of two complete cycles (see 3.3.6) at nominal fluid temperatures of 60 °F (16 °C), 104 °F (40 °C), and 130 °F (54 °C) and verified to meet the specified performance requirements (see 3.3.9). Total stroke time and time spent in the dashpot for each actuator stroke shall be recorded for information.

4.5.11 Maintainability demonstration. The maintainability requirement specified in 3.8 shall be demonstrated. The demonstration shall include the time required to accomplish normal corrective maintenance actions and the ease of actuator disassembly and reassembly. Failure to satisfy the MTTR specified in 3.8 shall be cause for rejection.

4.5.12 Vibration test. Actuators shall meet the type I environmental vibration criteria of MIL-STD-167-1 to ensure compliance with the requirements of 3.5.2. Actuators shall be tested in each of the three mutually perpendicular axes using the method specified (see 6.2).

4.5.13 Shock test. Actuators shall be qualified by test or extension in accordance with MIL-DTL-901 to ensure compliance with the requirements of 3.5.3. During testing, actuators shall be pressurized to the rated operating pressure. Electrical position switches may be simulated during shock testing. Post-shock test inspection shall include visual examination for evidence of physical damage as well as functional and operational tests to demonstrate all applicable performance requirements are met.

4.5.14 Endurance test. The actuator shall be endurance tested to the specified number of cycles at the rated actuator output and operating pressure. Unless otherwise specified (see 6.2), the test shall be conducted at a cycle rate of two to six cycles per minute. Periodically during the test, the actuator shall be examined for external leakage, cracks, or loose fasteners. Seals shall be replaced at a minimum of 15,000 cycles, or as specified (see 6.2). Any of these defects along with the inability to develop the specified output or premature seal replacement shall constitute a failure of the test. Upon completion of the endurance test, the group B tests in table I shall be repeated. The proof pressure test specified in 4.5.3 need not be repeated. Inability to meet the requirements specified in 3.3.6 shall constitute a failure of the test. Unless otherwise specified (see 6.2), hydraulic fluid used for the endurance test shall not be cleaner than the particle count limits of SAE AS4059 class 8 as verified immediately prior to conducting the endurance test.

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4.5.15 Salt fog test. Unless otherwise specified (see 6.2), the actuator shall be subjected to the salt fog test procedure specified in MIL-STD-810, method 509.7 for no less than four, alternating 24-hour periods (two wet and two dry) to ensure compliance with the requirements specified in 3.5.4.d. The salt concentration for the water shall be 5±1 percent. The actuator shall be set up in its normal operating mode. After completion of testing and cleaning, the base metal shall not be visible through the coating or finish, nor shall there be any evidence of corrosion (existence of any visible oxide), peeling, chipping, or blistering. Any evidence of these defects shall constitute a failure of the test. Approval by similarity may be extended to other actuators from the same manufacturer that use identical base and coating materials and equivalent coating thicknesses as the actuator tested.

4.5.16 Burst pressure verification. The actuator shall be verified to meet the burst pressure requirement specified in 3.3.1.3 by analysis or test. If verified by test, the actuator shall be subject to a static burst pressure test using either the specified test fluid (see 4.4.1) or water for a period not less than five minutes. Internal pressure boundary parts may be removed and other actuator ports capped. Burst pressure testing shall be the last test given the potentially destructive nature of the test. After test completion, the tested actuator shall be permanently marked as not acceptable for service.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The actuators covered in this specification are intended for use in ship hydraulic systems. Actuators are acceptable for use in systems using either petroleum-based, water-glycol, triaryl phosphate ester, or synthetic hydrocarbon hydraulic fluids.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Actuator type (see 1.2).
c. First article inspection requirements, if other than specified (see 3.1 and 6.3).
d. Rated operating pressure, if other than specified (see 3.3.1.1).
e. End-use hydraulic system fluid (see 3.3.2).
f. Fluid compatibility requirements, if other than specified (see 3.3.2).
g. Spatial envelope requirements (see 3.3.3).
h. Rotary actuator shaft rotating angle and allowable tolerance (see 3.3.5).
i. Minimum operating life, if other than specified (see 3.3.6).
j. Minimum and maximum actuator output (see 3.3.7).
k. Dashpot performance requirements when specified (see 3.3.9).
l. Internal leakage rate requirements, if other than specified (see 3.3.10).
m. Seal, packing, seal gland, and back-up ring design requirements, if other than specified (see 3.3.11).
n. Whether structureborne noise limits are required (see 3.3.12).
o. Allowable structureborne noise limits when required (see 3.3.12).
p. When brazing or welding is permitted (see 3.4.3).

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q. Seal gland chamfer design, if other than specified (see 3.4.4).
r. Actuator port requirements, if other than specified (see 3.4.5).
s. Fluid porting requirements, if other than specified (see 3.4.6).
t. Rotary actuator gear chamber fluid, if other than specified (see 3.4.9).
u. Rotary actuator spline fit (see 3.4.9.1).
v. When rotary actuators do not require a relief valve or other pressure limiting device (see 3.4.9.2).
w. Relief valve or pressure-limiting device operating requirements when required (see 3.4.9.2).
x. Position identification markings (see 3.4.10).
y. When remote electrical indication is required as part of the actuator assembly (see 3.4.10.1).
z. Sensitive switch requirements, if other than specified (see 3.4.10.1).
aa. Mounting interface configuration (see 3.4.12).
bb. Ambient pressure and temperature requirements, if other than specified (see 3.5.1).
cc. Shock grade and class (see 3.5.3).
dd. Nameplate requirements, if other than specified (see 3.7).
ee. Test fluid, if other than specified (see 4.4.1).
ff. Noise test conditions, if other than specified (see 4.5.9).
gg. Vibration test method (see 4.5.12).
hh. Endurance test cycle rate, if other than specified (see 4.5.14).
ii. Allowable seal replacements during endurance testing, if other than specified (see 4.5.14).
jj. Fluid cleanliness requirement prior to endurance testing, if other than specified (see 4.5.14).
kk. Salt fog test procedure, if other than specified (see 4.5.15).
ll. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, unless specific guidance is provided by the contracting officers to offerors (see 6.2), the first article item(s) may be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's inventory. The contracting officer should include specific instructions in the acquisition documents regarding the number of items to be tested (if more than one), arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Re-use of legacy hydraulic actuators (submarines only). A legacy hydraulic actuator is defined as any hydraulic actuator qualified in accordance with a previous revision of this specification for use on a past or present naval shipboard hydraulic system. In the event a legacy hydraulic actuator design is considered for re-use in a new or redesigned naval submarine hydraulic system for any reason, such as satisfying operational requirements, reducing component development costs, or supporting component commonality goals, that actuator design must first be verified to meet the minimum technical requirements provided in Appendix A.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.6 Subject term (key word) listing.

Fluid power

Hydraulic actuation

Hydraulic cylinder

Hydraulic fluid

Hydraulic system

Rack-and-pinion

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

MINIMUM TECHNICAL REQUIREMENTS FOR THE USE OF LEGACY ACTUATOR DESIGNS IN SUBMARINE HYDRAULIC SYSTEM APPLICATIONS

A.1 SCOPE

A.1.1 Scope. This appendix contains the minimum technical requirements to allow for the re-use of legacy hydraulic actuator designs in new or redesigned naval submarine hydraulic systems. Legacy actuator designs shall be verified to meet all requirements contained herein to be a permissible candidate for re-use. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance when re-use of legacy hydraulic actuators designs is considered.

A.2 DESIGN REQUIREMENTS

A.2.1 Material requirements. Actuator materials shall adhere to requirements specified in 3.2.

A.2.2 Pressure requirements.

A.2.2.1 Rated operating pressure. For legacy hydraulic actuator designs intended for use in Submarine Safety Design Review (SSDR) applications, external pressure boundary parts shall be designed to a minimum factor of safety of one using one-quarter of the minimum specified tensile strength or two-thirds of the minimum specified yield strength of the material, whichever is lower, when subjected to 110 percent of the rated operating pressure (see 3.3.1.1).

A.2.2.2 Proof pressure. Pressure containing parts shall be designed such that they do not experience a stress in excess of 90 percent of the minimum specified yield strength when subjected to proof pressure (see 3.3.1.2).

A.2.2.3 Burst pressure. Actuators shall be designed such that no part of the external pressure boundary ruptures due to excessive material deflection or failure when subjected to burst pressure (see 3.3.1.3).

A.2.3 Fastener requirements.

A.2.3.1 Preload torques. Preload torques specified for fasteners used in bolted joints subject to rated operating pressure (see 3.3.1.1) shall develop a clamping force that exceeds the applied hydraulic load at 110 of rated operating pressure (see 3.3.1.1). Specified preload torques shall not result in stresses in excess of the minimum material yield strength of the fastener.

A.2.3.2 Threaded inserts. Where threaded inserts are used, the minimum insert length shall not result in a stress exceeding the minimum ultimate tensile strength of the tapped material if the fastener is tightened to 135 of the maximum specified preload torque. Additionally, the specified preload torque range shall be such that the shear stress is limited to no more than two-thirds of the shear yield strength of the internal and external threads of the threaded insert at the minimum length of engagement.

A.2.4 Gear chamber requirements.

A.2.4.1 Gear chambers equipped with relief valves. No pressure containing part of a gear chamber equipped with a relief valve shall experience a stress in excess of 90 percent of the minimum specified yield strength of the material when subjected to 135 percent of the maximum relief valve set pressure (see 3.4.9.2). For legacy hydraulic actuator designs intended for use in SSDR applications, pressure containing parts of a gear chamber equipped with a relief valve shall be designed to require a minimum factor of safety of one-quarter of the minimum specified tensile strength or two-thirds of the minimum specified yield strength of the material, whichever is lower, when subjected to the maximum relief valve set pressure (see 3.4.9.2).

A.2.5 Castings and cast material requirements. Castings and cast materials shall be inspected as specified in 3.4.13.

A.2.6 Coating requirements. External surfaces of non-corrosion resistant metal, with the exception of threaded areas or machined surfaces in rolling or sliding contact, shall be cleaned, prepared, and coated as specified in 3.6.3.

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A.3 VERIFICATION

A.3.1 Legacy actuator design requirement verification. Design requirements of A.2.1, A.2.5, and A.2.6 shall be applied for any legacy actuator design under consideration for re-use in new or redesigned submarine hydraulic system applications. Design requirements of A.2.2 through A.2.4 shall be verified by analysis prior to re-use.

A.3.2 Conformance inspection. Conformance inspection shall be as specified in 4.3.

A.4 ADOPTION OF BEST PRACTICES

A.4.1 Best practices and design improvements. Legacy hydraulic actuator designs shall incorporate best engineering practices and design improvements to the maximum extent practical without requiring redesign of the legacy actuator(s) under consideration for re-use in new or redesigned submarine hydraulic system applications. These best practices and design improvements include, but are not limited to, the following:

a. Reduction or elimination of cadmium and zinc-plated parts;
b. Elimination of ester-based polyurethane seal compounds;
c. Reduction or elimination of welding and brazing;
d. Sufficient thread engagement to develop the ultimate tensile strength of the fastener as specified in 3.4.7;
e. Elimination of solid-type threaded inserts;
f. Adjustment of shim design to prevent loading cylinder sleeves in compression; and
g. Reduction or elimination of castings and cast materials.

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CONCLUDING MATERIAL

Preparing activity:
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